G11 PHYSICS

Assignment 4 Solutions

Total: 34 marks

Scenario. A ball is launched from a rooftop that is $H = 20\,\text{m}$ above the ground. It is launched at an angle of $\theta = 37°$ above the horizontal with an initial speed of $v_0 = 25\,\text{m/s}$. Take the launch point as the origin, with x pointing horizontally and y pointing vertically upward. Use $g = 10\,\text{m/s}^2$, $\sin 37° = 0.6$, and $\cos 37° = 0.8$.

Part A. Kinematics of the Launch *(15 marks)*

(a) **[3 marks]**

Resolving the initial velocity into components:

$$v_{0x} = v_0 \cos\theta = 25 \times 0.8 = 20\,\text{m/s}, \qquad v_{0y} = v_0 \sin\theta = 25 \times 0.6 = 15\,\text{m/s}.$$

Since $a_x = 0$ and $a_y = -g$, the kinematic equations give:

$$x(t) = v_{0x}\,t = 20t, \qquad y(t) = v_{0y}\,t - \tfrac{1}{2}gt^2 = 15t - 5t^2.$$

The full position vector is therefore

$$\boxed{\vec{s}(t) = 20t\,\hat{x} + \left(15t - 5t^2\right)\hat{y} \quad \text{(metres, } t \text{ in seconds).}}$$

(b) **[4 marks]**

At maximum height the vertical velocity is zero:

$$v_y(t) = \frac{dy}{dt} = 15 - 10t = 0 \implies \boxed{t^* = 1.5\,\text{s.}}$$

Substituting back into $y(t)$:

$$H_{\max} = y(t^*) = 15(1.5) - 5(1.5)^2 = 22.5 - 11.25 = \boxed{11.25\,\text{m above the launch point.}}$$

The ball therefore reaches $11.25 + 20 = 31.25\,\text{m}$ above the ground.

(c) **[5 marks]**

The ball lands when $y = -20\,\mathrm{m}$ (ground is $20\,\mathrm{m}$ below the launch point):

$$15t - 5t^2 = -20.$$

Rearranging:

$$5t^2 - 15t - 20 = 0 \implies t^2 - 3t - 4 = 0.$$

Factorising (or using the quadratic formula):

$$(t-4)(t+1) = 0 \implies t = 4\,\mathrm{s} \quad \text{or} \quad t = -1\,\mathrm{s}.$$

Accepted root: $\boxed{t_{\text{land}} = 4\,\mathrm{s}.}$

Rejected root: $t = -1\,\mathrm{s}$ is rejected because it corresponds to a time *before* the ball was launched ($t < 0$), which is not physically meaningful.

(d) **[3 marks]**

Using the landing time $t_{\text{land}} = 4\,\mathrm{s}$:

$$x_{\text{land}} = 20 \times 4 = \boxed{80\,\mathrm{m}.}$$

The ball lands $80\,\mathrm{m}$ horizontally from the base of the building.

Part B. Velocity, Speed & Acceleration *(11 marks)*

(a) **[5 marks]**

At $t = 4\,\mathrm{s}$:

$$v_x = 20\,\mathrm{m/s} \quad \text{(constant throughout)},$$
$$v_y = 15 - 10(4) = 15 - 40 = -25\,\mathrm{m/s}.$$

The velocity vector at impact is:

$$\vec{v} = 20\,\hat{x} - 25\,\hat{y} \quad \mathrm{m/s}.$$

Speed at impact:

$$|\vec{v}| = \sqrt{v_x^2 + v_y^2} = \sqrt{20^2 + 25^2} = \sqrt{400 + 625} = \sqrt{1025} \approx \boxed{32.0\,\mathrm{m/s}.}$$

Angle below the horizontal:

$$\alpha = \arctan\left(\frac{v_y}{v_x}\right) = \arctan\left(\frac{-25}{20}\right) = \arctan(-1.25) \approx \boxed{-51.3^\circ \text{ from the horizontal.}}$$

(b) **[4 marks]**

Physical reasoning. The speed of the projectile equals v_0 whenever the *magnitude* of the velocity vector returns to $25\,\text{m/s}$. Since v_x is constant, the speed depends only on $|v_y|$. The speed equals v_0 when $|v_y| = v_{0y} = 15\,\text{m/s}$, i.e. when the ball is at the same height as the launch point ($y = 0$). This happens at $t = 0$ (launch) and again on the way down when $y = 0$.
Algebraic verification. Setting $y(t) = 0$:

$$15t - 5t^2 = 0 \implies 5t(3 - t) = 0 \implies t = 0 \text{ or } t = 3\,\text{s}.$$

At $t = 3\,\text{s}$: $v_y = 15 - 10(3) = -15\,\text{m/s}$, so

$$|\vec{v}| = \sqrt{20^2 + 15^2} = \sqrt{625} = 25\,\text{m/s} = v_0. \checkmark$$

The speed equals v_0 at $t = 0\,\text{s}$ (launch) and $t = 3\,\text{s}$ (return to launch height).

(c) **[2 marks]**

Throughout the flight (air resistance neglected):

$$\vec{a}(t) = 0\,\hat{x} - g\,\hat{y} = -10\,\hat{y}\ \ \text{m/s}^2.$$

Its magnitude is $|\vec{a}| = g = 10\,\text{m/s}^2$, which is **constant and never zero**. Even at the highest point the acceleration is g downward (the ball's vertical velocity is instantaneously zero there, but the force of gravity (and hence the acceleration) acts continuously).

Part C. Integrals & Trajectory *(8 marks)*

(a) **[4 marks]**

With $v_y(t) = 15 - 10t$ and $t^* = 1.5\,\text{s}$:

$$\int_0^{t^*} v_y(t)\,dt = \int_0^{1.5} (15 - 10t)\,dt = \Big[15t - 5t^2\Big]_0^{1.5} = \big(22.5 - 11.25\big) - 0 = \boxed{11.25\,\text{m.}}$$

Geometric interpretation. The integral equals the *area under the v_y–t graph* from $t = 0$ to $t = 1.5\,\text{s}$. Since v_y is a straight line from 15 to 0, this area is a triangle: $\frac{1}{2} \times 1.5 \times 15 = 11.25\,\text{m}$.

Physical interpretation. The result is the net *vertical displacement* of the ball from launch until it reaches its maximum height — consistent with $H_{\text{max}} = 11.25\,\text{m}$ found in Part A(b).

(b) **[4 marks]**

From Part A(a): $x = 20t$, so $t = \dfrac{x}{20}$. Substituting into $y(t)$:

$$y = 15 \cdot \frac{x}{20} - 5\left(\frac{x}{20}\right)^2 = \frac{3x}{4} - 5 \cdot \frac{x^2}{400} = \frac{3x}{4} - \frac{x^2}{80}.$$

Rewriting:

$$\boxed{y = \frac{3}{4}x - \frac{1}{80}x^2.}$$

This is of the form $y = Ax + Bx^2$ with $B = -\dfrac{1}{80} < 0$, confirming a **downward-opening parabola**. The coefficient of x^2 is $-\dfrac{1}{80}$.

This matches the general trajectory formula $y = x\tan\theta - \dfrac{g}{2v_0^2\cos^2\theta}x^2$ with $\tan 37^\circ = 0.75$ and $\dfrac{g}{2v_0^2\cos^2\theta} = \dfrac{10}{2(625)(0.64)} = \dfrac{10}{800} = \dfrac{1}{80}$. ✓

End of solutions.